

Annotator Disagreement, Not Spatial Context, Bounds Pulmonary Nodule Malignancy-Risk Classification on LIDC-IDRI

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Abstract

Deep learning studies on LIDC-IDRI routinely report that richer spatial context—moving from a single slice to a stack of slices to a full volume—improves pulmonary nodule malignancy-risk classification. We test this directly under a single controlled pipeline in which representation is the only variable, and find no such benefit. Across 2D (one slice), 2.5D (five slices) and 3D (full volume) encoders sharing identical preprocessing, patient-level splits, loss, optimiser, augmentation, model-selection rule and metrics, all representations converge to a three-class macro-F1 of approximately 0.60. Self-supervised MoCo v2 pretraining on 5133 unlabelled training-patient nodules produced no measurable gain in the 2D and 2.5D arms, and an apparent gain only in the 3D arm, where validation did not reliably rank configurations. We show instead that performance is bounded by *annotator disagreement*. On a binary benign-versus-malignant cohort drawn from the same images and patient splits, accuracy is 0.945 and AUC 0.987 on nodules with concordant radiologist labels, but falls to 0.805 and 0.902 on the remainder; classification accuracy tracks the median malignancy rating, collapsing from ~ 0.90 at the rating extremes to 0.667 at a median of 3.5, a value that is not an original radiologist rating but an artefact of averaging disagreeing readers. Models are appropriately uncertain rather than confidently wrong, with mean confidence 0.762 on correct predictions against 0.462 on errors. We argue that a substantial part of the headroom commonly attributed to architecture on this dataset is instead label noise, and that LIDC-IDRI results should be reported stratified by annotator agreement.

1 Introduction

Computed tomography is volumetric, and a long line of work argues that deep models should exploit that structure [9, 14, 17, 19]: 2.5D models stack neighbouring slices, and 3D models consume whole volumes. Reported gains are usually modest and are almost always confounded, because the compared systems differ not only in spatial context but in preprocessing, intensity windowing, augmentation, capacity, training schedule and evaluation protocol.

We ask a narrower question. *If everything except the input representation is held fixed, how much does spatial context actually contribute?* We build one pipeline in which 2D, 2.5D and 3D arms share the same nodule crops, the same Hounsfield-unit window, the same patient-level splits, the same class-weighted loss, the same optimiser and schedule, the same augmentation recipe, the same early-stopping criterion and the same metric code. The encoders differ only in input dimensionality; the 2D and 2.5D models are the same ResNet-18 differing in a single argument, `in_channels`.

Our contributions:

1. Under this controlled comparison, added spatial context yields no improvement. Three-class macro-F1 is 0.6059 ± 0.024 (2D), 0.5785 ± 0.022 (2.5D) and 0.5653 ± 0.041 (3D), all within one to two seed standard deviations of one another.

2. MoCo v2 pretraining on 5133 unlabelled training-patient nodules gives no measurable benefit in the 2D or 2.5D arms despite the pretext task itself converging well. It does improve the 3D arm by +0.043 macro-F1, but 3D validation ranked configurations inconsistently with test, so we treat this as an observation requiring more seeds rather than a result.
3. We locate the performance ceiling in annotator disagreement rather than model capacity, and quantify it: binary accuracy is 0.945 / AUC 0.987 on concordantly labelled nodules versus 0.805 / 0.902 on the remainder.
4. We show that augmentation does not reduce memorisation, and report the seed-variance reduction it does deliver.

2 Related Work

2D, 2.5D and 3D nodule classification. Early deep models for LIDC-IDRI malignancy classified 2D patches at several scales or crops [15, 16]. Multi-view 2D networks, which aggregate several planes through a nodule, were introduced for false positive reduction [14], and fully 3D networks followed for malignancy [9, 19]. ACS convolutions [17] reuse 2D-pretrained kernels on 3D input and report gains over both pure 2D and 3D baselines on LIDC-IDRI. These studies generally compare systems that differ in more than input dimensionality, and many drop nodules with an indeterminate rating, which changes both the task and its difficulty. Our contribution is not a new architecture but a comparison in which only the representation varies.

Label selection and ambiguity on LIDC-IDRI. Baltatzis et al. [3] showed that the choice of which LIDC-IDRI nodules to include, and how reader ratings are aggregated, can move reported performance by more than the differences between methods. Our ceiling analysis is complementary: rather than varying the selection rule, we hold one rule fixed for training and stratify the *evaluation* by reader agreement, and we do this across three input representations. Multiple LIDC-IDRI annotators have also been used to model ambiguity directly, for example in segmentation [11], and reader disagreement has been treated as a prediction target in its own right [13]. More broadly, label noise is a recognised problem in medical imaging [10].

Self-supervised pretraining. Contrastive pretraining such as MoCo [8] has been shown to help medical image classification when the pretraining pool is large [2], and Models Genesis [18] provides 3D self-supervised encoders for CT. Our pool of 5133 nodules is small by these standards, and our results should be read as a test of in-domain, small-pool pretraining rather than of self-supervision in general.

3 Data

We use LIDC-IDRI [1], obtained through TCIA [5]. Consensus nodules were extracted as $104 \times 72 \times 80$ crops resampled to 1 mm isotropic spacing, with the nodule at the geometric centre of every crop; the measured mask centre-of-mass across random samples is $z \in [51.1, 52.1]$, $y \in [35.2, 35.7]$, $x \in [39.3, 39.9]$.

Labels. Each nodule carries malignancy ratings from up to four radiologists on a 1–5 scale. We use the project three-class rule: median 1.0–2.0 is *benign*, 2.5–3.0 *indeterminate*, and 3.5–5.0 *malignant*.

Crucially, medians of 2.5 and 3.5 are *not* radiologist ratings; they arise only when readers disagree. 449 nodules carry such midpoint medians, and we return to them in Section 5.4.

Cohorts. The labelled cohort is 2654 nodules (873 benign, 1226 indeterminate, 555 malignant), split at patient level into 1876 / 382 / 396 nodules over 602 / 134 / 134 patients. A binary cohort excluding *indeterminate* gives 1000 / 219 / 209 nodules; it is a strict subset of the same splits, so no patient moves between partitions. The self-supervised pool is 5133 nodules from 692 training-split patients, used without labels; it may include labelled training nodules but contains no validation or test patient.

Leakage control. We verified programmatically, and re-assert on every training run, that no patient and no nodule appears in more than one split, and that the self-supervised pool contains no validation or test patient. The check raises rather than warns.

4 Method

Preprocessing (shared). Stored volumes are raw Hounsfield units with observed values from -3024 to $+3080$. We clip to $[-1000, 400]$, scale to $[0, 1]$, then standardise with training-split statistics. This window is applied identically in all arms; it is the single most important shared choice, since an unstated difference here would masquerade as a representation effect.

Representations.

- **2D:** the central slice, (1, 72, 80), ResNet-18 [7].
- **2.5D:** five neighbouring slices as channels, (5, 72, 80), the same ResNet-18 with `in_channels=5`.
- **3D:** the full crop, (1, 104, 72, 80), a 3D ResNet-10 [6].

The 2D and 2.5D models differ by 12,544 parameters, entirely in the first convolution. We use ResNet-10 rather than ResNet-18 in 3D because a 3D ResNet-18 would carry roughly 33M parameters against 1876 training volumes.

Training (shared). AdamW [12], learning rate 10^{-3} , weight decay 0.05, cosine schedule, batch 64 (16 in 3D), dropout 0.5, label smoothing 0.10, inverse-frequency class weights, early stopping with patience 20 on validation macro-F1, maximum 80 epochs. Learning rate, weight decay, dropout and label smoothing were selected on validation only, by a single-seed sweep of eight configurations in the 2D arm, and then frozen for every other arm. Augmentation strength is not tuned; it is an experimental factor (Models A–C below). The test split was never used for tuning or checkpoint selection.

Models. **A:** supervised from random initialisation, no augmentation. **B:** MoCo v2 pretraining, then supervised fine-tuning without augmentation. **C:** as A, with the mild augmentation below. The binary runs use the Model C recipe.

Table 1: Test performance by representation, mean $\pm$ std over three seeds. All arms share one pipeline; only the representation and encoder differ.

Model	Metric	2D	2.5D	3D
A (supervised)	macro-F1	0.6059 ± 0.024	0.5785 ± 0.022	0.5653 ± 0.041
	accuracy	0.6002	0.5690	0.5564
	ROC-AUC	0.7565	0.7405	0.7223
B (MoCo)	macro-F1	0.5957 ± 0.009	0.5875 ± 0.011	0.6083 ± 0.008
	ROC-AUC	0.7579	0.7368	0.7373
C (augmented)	macro-F1	0.6134 ± 0.008	0.5934 ± 0.016	0.5738 ± 0.013
	ROC-AUC	0.7595	0.7340	0.7197
Binary	macro-F1	0.8227 ± 0.002	0.8047 ± 0.006	0.7866 ± 0.031
	ROC-AUC	0.8644	0.8612	0.8443

Augmentation. Training only, Model C and binary runs. Horizontal and vertical flips ($p = 0.5$ each), a combined affine transform ($p = 0.8$; rotation $\pm 15^\circ$, translation $\pm 6\%$, scale 0.9–1.1), intensity jitter ($p = 0.5$; multiplicative 0.9–1.1 then additive ± 0.05) and Gaussian noise ($p = 0.3$, $\sigma = 0.02$). A *strong* variant, reported only as an ablation, doubles the geometric ranges, triples the intensity ranges, uses $\sigma = 0.05$ and adds random erasing ($p = 0.25$). In 3D, flips are in-plane only, so the volume is never flipped along z . In the reported 3D runs the $\pm 15^\circ$ rotation was applied in the sagittal (y – z) plane rather than the axial plane, so 3D augmentation is not an exact analogue of the 2D recipe (see Limitations).

Self-supervision. MoCo v2 [4, 8] on the 5133-nodule pool, no labels: queue $K = 4096$, temperature 0.2, momentum 0.999, two-layer MLP head, SGD. In 2D and 2.5D we pretrain for 200 epochs at batch 128 and learning rate 0.015; in 3D we pretrain for 100 epochs at batch 32 and learning rate 0.004, with views drawn from the strong 3D augmentation. With a single device we approximate shuffle-BN by permuting the key batch and forwarding it in chunks, so each chunk receives its own batch statistics.

Evaluation. Accuracy, balanced accuracy, macro and per-class precision / recall / F1, confusion matrices, and one-vs-rest ROC-AUC. Every figure is the mean $\pm$ standard deviation over three seeds, with percentile bootstrap confidence intervals on test macro-F1. Majority-class baselines are macro-F1 0.2138 (three-class) and 0.3589 (binary).

5 Results

5.1 Spatial context does not help

Table 1 and Figure 1 show no benefit from additional spatial context. Moving from one slice to five changes three-class macro-F1 by -0.027 (Model A) and -0.020 (Model C); extending to the full 104-slice volume changes it by -0.041 and -0.040 respectively. With per-seed bootstrap intervals spanning roughly ± 0.05 on a 396-nodule test set, none of these are distinguishable from zero, and all are negative in sign. The strongest model overall is 2D (Model C, 0.6134); the strongest 3D model (Model B) reaches 0.6083, an indistinguishable figure that required a 327-minute pretraining

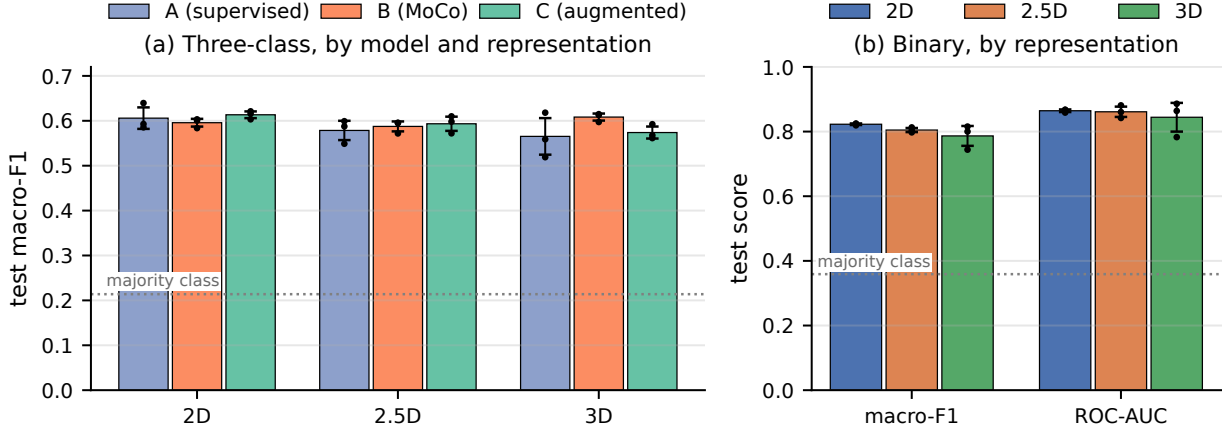


Figure 1: Test performance by representation. Bars are means over three seeds, error bars one standard deviation, dots individual seeds; the dotted line is the majority-class baseline. (a) Three-class macro-F1 for Models A–C. (b) Binary macro-F1 and ROC-AUC. No representation separates from the others.

Table 2: Per-class test F1 (three-class), mean over three seeds. Malignant is the easiest class and benign the hardest in every arm.

Model	Class	2D	2.5D	3D
A (supervised)	benign	0.5362	0.5328	0.4583
	indeterminate	0.6001	0.5550	0.5733
	malignant	0.6814	0.6477	0.6643
B (MoCo)	benign	0.5169	0.4906	0.4860
	indeterminate	0.6080	0.5801	0.6315
	malignant	0.6623	0.6917	0.7074
C (augmented)	benign	0.5421	0.5228	0.4436
	indeterminate	0.6274	0.5853	0.5809
	malignant	0.6708	0.6720	0.6970

run plus 31.7 minutes of fine-tuning per seed on a Kaggle GPU, against 2.4 minutes per seed for 2D Model C on an Apple M2. Because the arms ran on different hardware these times are not a controlled cost comparison, but they put 3D more than an order of magnitude above 2D (Table 3, Figure 2). Binary ROC-AUC is nearly identical across representations (0.8644, 0.8612, 0.8443), indicating that ranking quality—not merely the decision threshold—is unchanged. Per-class F1 (Table 2) follows the same ordering in every arm: malignant is easiest and benign hardest, despite benign being the larger training class.

5.2 Self-supervised pretraining helps only the 3D encoder

MoCo v2 changed three-class test macro-F1 by -0.0102 in 2D and $+0.0090$ in 2.5D: opposite signs, both inside the seed spread. ROC-AUC was essentially unchanged in 2D (0.7579 vs 0.7565). The pretext task itself trained well, with InfoNCE loss falling from 7.48 to 4.14 and instance-discrimination accuracy reaching 0.863 over 200 epochs (Figure 3), so in these two representations

Table 3: Computational cost. Supervised minutes are per seed for Models A / B / C; MoCo pretraining is a single run shared by the three Model B seeds. Times were measured on different hardware and are not directly comparable across arms.

Arm	Encoder	Parameters	Supervised min/seed	MoCo min	Hardware
2D	ResNet-18	11,171,779	2.1 / 2.2 / 2.4	57.0	Apple M2
2.5D	ResNet-18	11,184,323	3.2 / 3.0 / 3.4	35.2	Apple M2
3D	3D ResNet-10	14,357,059	24.5 / 31.7 / 42.7	327.1	Kaggle GPU

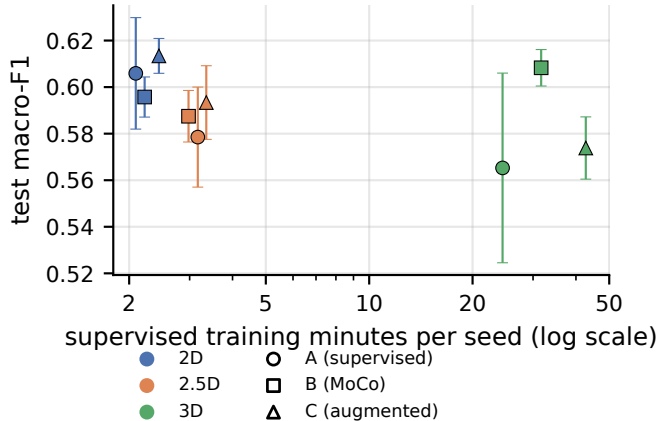


Figure 2: Test macro-F1 against supervised training time per seed (log scale). Error bars are one standard deviation over seeds. 3D costs more than an order of magnitude more time without a corresponding gain.

the outcome is a transfer failure rather than an optimisation failure. Instance discrimination rewards features that identify a crop—surrounding vasculature, pleural contact, framing—which need not be the features that indicate malignancy risk.

The 3D arm behaves differently. There MoCo gives +0.043 macro-F1 over the supervised baseline (0.6083 vs 0.5653) and +0.035 over the augmented model (0.5738), making the pretrained model the strongest 3D configuration we obtained. The pattern across representations is consistent with pretraining mattering in proportion to how data-starved supervised training is: the 3D encoder carries 14.4M parameters against 1876 training volumes, where the 2D encoder sees the same patients as 11.2M parameters over far simpler inputs.

We report this as an observation rather than a claim. In the 3D arm, validation macro-F1 ranked the three configurations differently from test: validation ordered them A (0.5698), B (0.5662), C (0.5375), whereas test ordered them B, C, A. Model B scored below Model A on validation yet 0.043 above it on test, so with three seeds we cannot separate a genuine pretraining benefit from checkpoint-selection noise. Distinguishing the two requires more seeds than we ran.

5.3 Augmentation does not reduce memorisation

On test, mild augmentation gave the highest 2D score (0.6134 vs 0.6059 unaugmented) and strong augmentation the lowest (0.5908). These differences are inside the seed spread, and validation did not reproduce the ordering (0.5982, 0.5962 and 0.5823 for none, mild and strong), so we do not

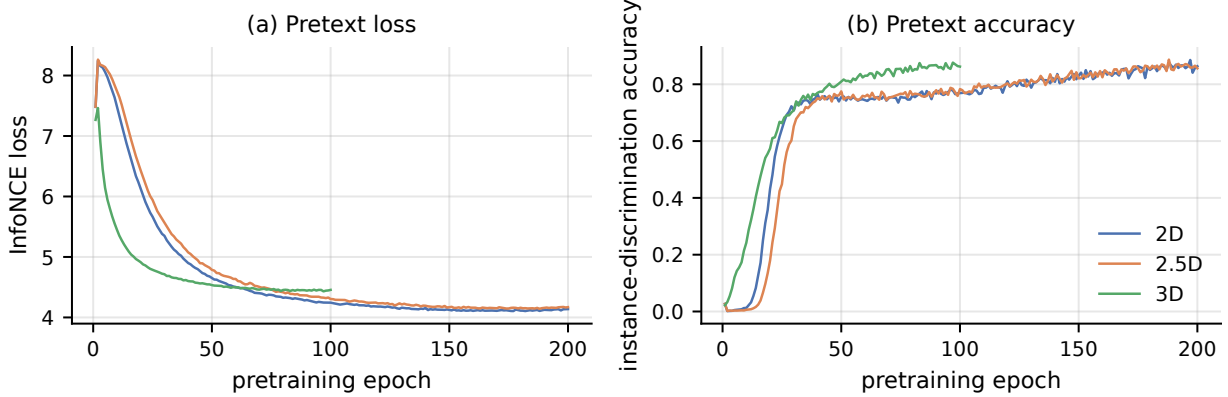


Figure 3: MoCo v2 pretext training on the 5133-nodule pool. All three encoders learn the pretext task; 3D uses a 100-epoch schedule.

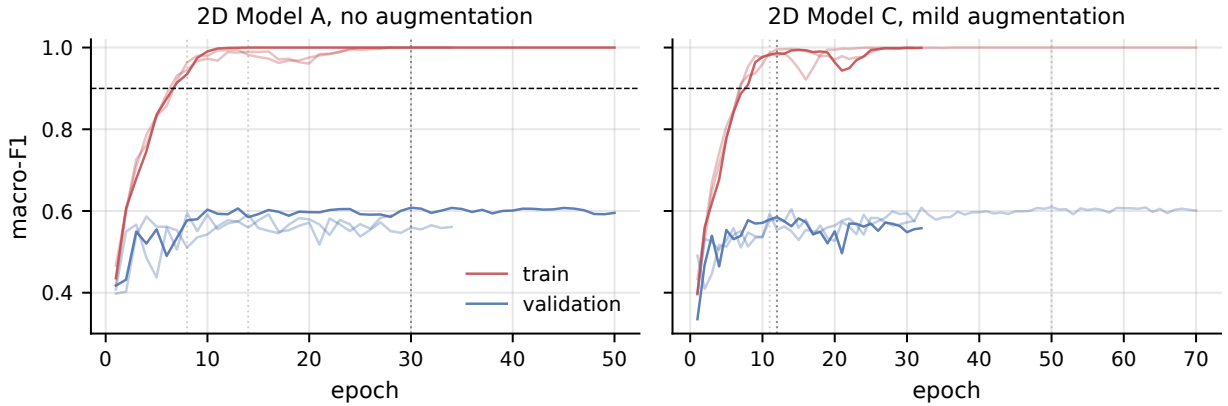


Figure 4: 2D training and validation macro-F1 per epoch, without and with mild augmentation. Seed 0 is drawn solid, seeds 1–2 faded; dotted verticals mark the selected epochs and the dashed line marks 0.90. Training macro-F1 passes 0.90 within 7–8 epochs either way while validation plateaus near 0.6.

claim an optimal strength. The train–validation macro-F1 gap at the selected epoch (Figure 4) was unchanged: $+0.386$ without augmentation, $+0.394$ with mild augmentation. Peak 2D training macro-F1 reached 1.000 with and without mild augmentation (0.999 with strong augmentation), and training macro-F1 crossed 0.90 at epoch 7–8 regardless of augmentation. What augmentation reliably delivered was *stability*: the seed standard deviation of test macro-F1 fell from ± 0.024 to ± 0.008 . Early stopping, not augmentation, is what limits overfitting here.

5.4 The ceiling is annotator disagreement

Table 4 and Figure 5 are our central result. On nodules where radiologists agreed, a single-slice ResNet-18 reaches 0.945 accuracy and 0.987 AUC. The pooled accuracy of the same seed-averaged predictions, 0.842 (176/209), is depressed almost entirely by nodules where they did not.

Accuracy tracks the median rating directly: 0.898 at median 1.0, 0.905 at 2.0, 0.900 at 5.0—but 0.667 at 3.5 and 0.697 at 4.0. Recall that 3.5 is not a rating any radiologist assigned. Nodules

Table 4: Binary test performance stratified by label confidence (2D, probabilities averaged over three seeds). “Confident” means no midpoint median, at least three readers, and no disagreement flag.

Subset	n	Accuracy	AUC
Confident labels	55	0.945	0.987
Remainder	154	0.805	0.902

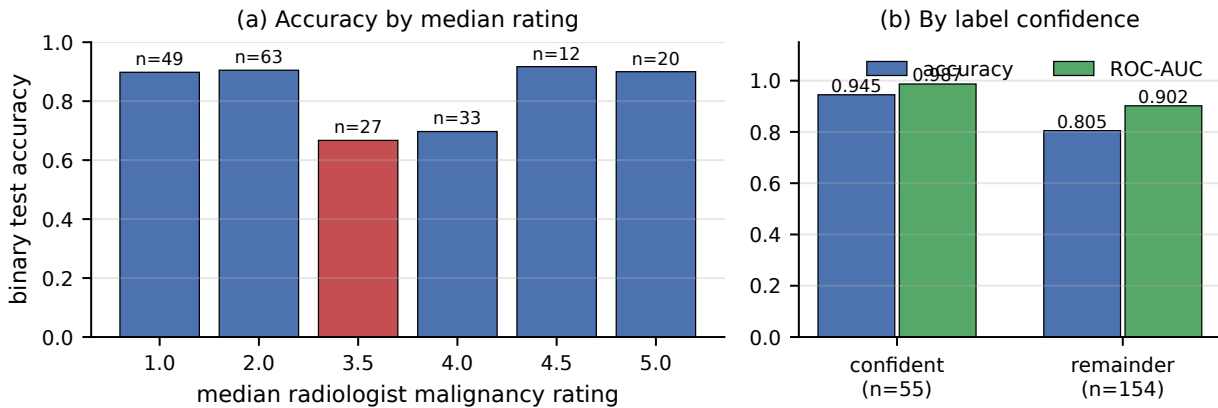


Figure 5: Binary test accuracy against label confidence (2D, probabilities averaged over three seeds). (a) By median radiologist rating; the red bar is a midpoint median (3.5), which no radiologist assigned. (b) By the confidence subsets of Table 4.

flagged as midpoint medians score 0.667 against 0.868 for the rest; single-reader nodules score 0.722 against 0.903 for three-reader nodules.

The three-class confusion matrices (Figure 6) show the same structure. Pooling over seeds for 2D Model C, benign $\leftrightarrow$ indeterminate accounts for 284 errors and indeterminate $\leftrightarrow$ malignant for 135, while benign $\leftrightarrow$ malignant accounts for only 41. The 2.5D (322 / 133 / 38) and 3D (359 / 117 / 40) arms follow the same pattern. The clinically meaningful extremes are separable; the boundary that is not is the one defined by reader disagreement.

Finally, the models are markedly less confident when they are wrong: mean confidence is 0.762 on correct predictions and 0.462 on errors, and 19 of 33 binary errors fall at median 3.5 or 4.0.

6 Discussion

Three architectures spanning one to 104 slices, each additionally pretrained on 5133 unlabelled volumes, converge to the same three-class performance: the best configuration of every representation lands between 0.5934 and 0.6134 macro-F1. The most parsimonious explanation is that they are all saturating the information available in the labels rather than in the images. On that reading, pretraining helps the 3D encoder not by supplying information the images lacked but by partially compensating for the optimisation disadvantage of fitting 14.4M parameters to 1876 volumes—it moves the weakest representation up to the common ceiling rather than through it.

This has a practical consequence for the literature. Improvements of one or two macro-F1 points on LIDC-IDRI, reported without stratification by annotator agreement and without confidence intervals, are difficult to distinguish from noise: on a 396-nodule test set our bootstrap intervals

			2D, Model C			2.5D, Model C			3D, Model C		
true	benign		193 55%	139 40%	19 5%	198 56%	138 39%	15 4%	161 46%	175 50%	15 4%
	indet.		145 26%	356 63%	60 11%	184 33%	321 57%	56 10%	184 33%	330 59%	47 8%
	malig.		22 8%	75 27%	179 65%	23 8%	77 28%	176 64%	25 9%	70 25%	181 66%
			benign	indet.	malig.	benign	indet.	malig.	benign	indet.	malig.
			predicted			predicted			predicted		

Figure 6: Three-class test confusion matrices for Model C, pooled over three seeds (counts and row percentages). Errors concentrate on the benign/indeterminate and indeterminate/malignant boundaries in every arm.

span ± 0.05 . We recommend that LIDC-IDRI results be reported separately for unanimously and ambiguously labelled nodules, and that midpoint medians—which are averaging artefacts, not clinical categories—be flagged explicitly.

7 Limitations

The labelled cohort is small (1876 training nodules) relative to encoder capacity, and all models overfit substantially. Confidence intervals are correspondingly wide, and we have deliberately refrained from claiming that any representation is superior. Labels are radiologist opinion rather than histopathology, so the “ceiling” we measure is agreement with readers, not with ground truth. We evaluated a single self-supervised configuration per representation, and the 3D one used a shorter schedule than 2D and 2.5D; a negative transfer result for one MoCo v2 setting is not evidence that self-supervision cannot help. In the reported 3D runs that use augmentation (Model C, binary, and the MoCo views behind Model B), rotation was applied in the sagittal rather than the axial plane, so 3D augmentation is not an exact analogue of the 2D recipe; Model A is unaffected. Training times were measured on different hardware across arms. Each configuration was run with three seeds; this is enough to show that the representations do not separate, but not enough to resolve the $+0.043$ pretraining effect in the 3D arm, particularly as 3D validation macro-F1 ranked configurations inconsistently with test. The z axis was resampled from 2.5 mm, so inter-slice context is partly interpolated, which disadvantages the 2.5D and 3D arms to an unknown degree. Hyperparameters were tuned on a 382-nodule validation split, itself a small sample, and the search was conducted on the 2D arm and transferred unchanged to 2.5D and 3D. Holding hyperparameters fixed isolates the effect of the representation, but it does not give each architecture its best achievable configuration, and a per-arm search could narrow or widen the gaps we report.

8 Conclusion

Under a controlled single-pipeline comparison, additional spatial context did not improve pulmonary nodule malignancy-risk classification on LIDC-IDRI, and self-supervised pretraining improved only the representation that was furthest behind, without lifting it past the others. Performance is instead bounded by annotator disagreement: where radiologists agree, a single CT slice suffices for 0.945 accuracy and 0.987 AUC. We suggest that progress on this benchmark be measured against agreement-stratified baselines.

Code and reproducibility. All code, per-seed metrics, training curves and confusion matrices are available at <https://github.com/naazzarov/CT-2D-2.5D-3D-DeepLearning>.

References

- [1] Samuel G Armato III, Geoffrey McLennan, Luc Bidaut, Michael F McNitt-Gray, Charles R Meyer, Anthony P Reeves, et al. The lung image database consortium (lidc) and image database resource initiative (idri): a completed reference database of lung nodules on ct scans. *Medical Physics*, 38(2):915–931, 2011.
- [2] Shekoofeh Azizi, Basil Mustafa, Fiona Ryan, Zachary Beaver, Jan Freyberg, Jonathan Deaton, Aaron Loh, Alan Karthikesalingam, Simon Kornblith, Ting Chen, Vivek Natarajan, and Mohammad Norouzi. Big self-supervised models advance medical image classification. In *ICCV*, pages 3478–3488, 2021.
- [3] Vasileios Baltatzis, Kyriaki-Margarita Bintsi, Loïc Le Folgoc, Octavio E. Martinez Manzanera, Sam Ellis, Arjun Nair, Sujal Desai, Ben Glocker, and Julia A. Schnabel. The pitfalls of sample selection: a case study on lung nodule classification. In *Predictive Intelligence in Medicine (PRIME), MICCAI Workshop*, pages 201–211, 2021. arXiv:2108.05386.
- [4] Xinlei Chen, Haoqi Fan, Ross Girshick, and Kaiming He. Improved baselines with momentum contrastive learning. *arXiv preprint arXiv:2003.04297*, 2020.
- [5] Kenneth Clark, Bruce Vendt, Kirk Smith, John Freymann, Justin Kirby, Paul Koppel, et al. The cancer imaging archive (tcia): maintaining and operating a public information repository. *Journal of Digital Imaging*, 26(6):1045–1057, 2013.
- [6] Kensho Hara, Hirokatsu Kataoka, and Yutaka Satoh. Can spatiotemporal 3D CNNs retrace the history of 2D CNNs and ImageNet? In *CVPR*, pages 6546–6555, 2018.
- [7] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image recognition. In *CVPR*, pages 770–778, 2016.
- [8] Kaiming He, Haoqi Fan, Yuxin Wu, Saining Xie, and Ross Girshick. Momentum contrast for unsupervised visual representation learning. In *CVPR*, pages 9729–9738, 2020.
- [9] Sarfaraz Hussein, Kunlin Cao, Qi Song, and Ulas Bagci. Risk stratification of lung nodules using 3D CNN-based multi-task learning. In *Information Processing in Medical Imaging (IPMI)*, volume 10265 of *LNCIS*, pages 249–260, 2017.

- [10] Davood Karimi, Haoran Dou, Simon K. Warfield, and Ali Gholipour. Deep learning with noisy labels: exploring techniques and remedies in medical image analysis. *Medical Image Analysis*, 65:101759, 2020.
- [11] Simon A. A. Kohl, Bernardino Romera-Paredes, Clemens Meyer, Jeffrey De Fauw, Joseph R. Ledsam, Klaus H. Maier-Hein, S. M. Ali Eslami, Danilo Jimenez Rezende, and Olaf Ronneberger. A probabilistic U-Net for segmentation of ambiguous images. In *NeurIPS*, 2018.
- [12] Ilya Loshchilov and Frank Hutter. Decoupled weight decay regularization. In *ICLR*, 2019.
- [13] Maithra Raghu, Katy Blumer, Rory Sayres, Ziad Obermeyer, Bobby Kleinberg, Sendhil Mullainathan, and Jon Kleinberg. Direct uncertainty prediction for medical second opinions. In *ICML*, volume 97 of *PMLR*, pages 5281–5290, 2019.
- [14] Arnaud Arindra Adiyoso Setio, Francesco Ciompi, Geert Litjens, Paul Gerke, Colin Jacobs, Sarah J. van Riel, Mathilde Marie Winkler Wille, Matiullah Naqibullah, Clara I. Sánchez, and Bram van Ginneken. Pulmonary nodule detection in CT images: false positive reduction using multi-view convolutional networks. *IEEE Transactions on Medical Imaging*, 35(5):1160–1169, 2016.
- [15] Wei Shen, Mu Zhou, Feng Yang, Caiyun Yang, and Jie Tian. Multi-scale convolutional neural networks for lung nodule classification. In *Information Processing in Medical Imaging (IPMI)*, volume 9123 of *LNCS*, pages 588–599, 2015.
- [16] Wei Shen, Mu Zhou, Feng Yang, Dongdong Yu, Di Dong, Caiyun Yang, Yali Zang, and Jie Tian. Multi-crop convolutional neural networks for lung nodule malignancy suspiciousness classification. *Pattern Recognition*, 61:663–673, 2017.
- [17] Jiancheng Yang, Xiaoyang Huang, Yi He, Jingwei Xu, Canqian Yang, Guozheng Xu, and Bingbing Ni. Reinventing 2D convolutions for 3D images. *IEEE Journal of Biomedical and Health Informatics*, 25(8):3009–3018, 2021.
- [18] Zongwei Zhou, Vatsal Sodha, Md Mahfuzur Rahman Siddiquee, Ruibin Feng, Nima Tajbakhsh, Michael B. Gotway, and Jianming Liang. Models genesis: Generic autodidactic models for 3D medical image analysis. In *MICCAI*, volume 11767 of *LNCS*, pages 384–393, 2019.
- [19] Wentao Zhu, Chaochun Liu, Wei Fan, and Xiaohui Xie. DeepLung: Deep 3D dual path nets for automated pulmonary nodule detection and classification. In *IEEE Winter Conference on Applications of Computer Vision (WACV)*, pages 673–681, 2018.